



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

CIE - Imp - III
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USN

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	27 th January 2025	Maximum Marks	10 + 50
Course Code	HS351TA	Duration	120 Mins
Sem	V Semester	Improvement CIE	
ENTREPRENEURSHIP AND INTELLECTUAL PROPERTY RIGHTS			

Note:

1. Answer all the Questions.

PART - A

Sl. No.	Questions	M	BT	CO
1	_____ is a fundamental framework in marketing that helps businesses identify and reach their most valuable customers.	1	1	2
2	Creating a _____ is about identifying and clearly communicating what makes a product, service, or brand distinctively valuable to its target audience	1	1	4
3	A company is launching a premium product with a production cost of Rs 30 per unit. They estimate fixed costs for branding and marketing at Rs 2,00,000. The company plans to sell 10,000 units in the first year and targets a 40% profit margin. What should be the selling price per unit to achieve the desired profit margin?	2	2	2
4	Toyota developed the Toyota Way, a set of principles that focuses on _____ and respect for people.	1	1	4
5	Differentiate between Planning and Strategy.	2	1	2
6	With _____ growth, a company expands through its own operations using its own internal resources.	1	1	4
7	If a company's gross profit margin is increasing but its net profit margin is declining, what could be the reason?	2	2	3

PART - B

Sl. No.	Questions	M	BT	CO
1.	A tech company is developing a mobile app that combines fitness tracking with AI-based personalized coaching. They are seeking funding from venture capitalists. How can the company effectively articulate its unique value proposition in the business plan to stand out from competitors in the crowded fitness tech market?	10	4	4
2.	a) A start-up has developed a patent-pending water purification device that uses renewable energy. They aim to market it in developing countries. How should they address intellectual property, innovation, and scalability in their business plan to appeal to investors? b) A global e-commerce platform is considering a strategic alliance with a logistics company to enhance delivery efficiency in underserved regions. The e-commerce platform wants to improve delivery times, and the logistics company seeks access to a broader customer base. What factors should both companies consider when forming a strategic alliance, and how can they ensure that the partnership is mutually beneficial?	5 5	4 3	4 4
3.	What are the objectives of Financial Management? Investigate Financial Management Tools and Techniques.	10	2	2
4.	Illustrate how successful companies have leveraged each of the 4P	10	3	2



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5.	A manufacturing company needs Rs 20 Lacs to purchase advanced machinery. They are considering two options: • Taking a bank loan with a 7% annual interest rate. • Selling 20% of the company's equity to a private investor. 1. What factors should the company consider when choosing between equity and debt financing? Which option might be better if the company expects rapid growth? Justify with a case study. 2. What steps can the company take to manage debt repayment and avoid defaulting on the loan? 3. Illustrate how successful companies have leveraged each of the 4Ps to achieve market success.	10	4
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BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4
	Max Marks	Quiz	--	5	2	3	--	06	04	--	--
		Test	--	20	10	20	--	--	10	15	25



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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	27/01/2025	Maximum Marks	10 + 50
Course Code	IM352IA	Duration	30 + 90 Min
Sem	V	CIE – III	
OPERATIONS MANAGEMENT			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO																		
Part – A																						
1	A _____ is any resource whose capacity is less than or equal to the demand placed on it.	1	1	3																		
2	In product mix decisions, TOC prioritizes products based on _____ per unit of the bottleneck resource.	1	1	3																		
3	The Theory of Constraints uses the _____ process to repeat improvements once a bottleneck is resolved.	1	1	3																		
4	In manufacturing, reducing _____ time at a bottleneck can significantly improve efficiency.	1	1	3																		
5	Differentiate between Static Schedule and Dynamic Schedule	2	2	4																		
6	What is bottleneck scheduling, and why is it critical?	2	2	4																		
7	Using the Earliest Due Date (EDD) rule, determine the sequence for the following jobs based on their due dates: <table><tr><th>Job</th><th>Processing time(hr)</th><th>Due date(hr)</th></tr><tr><td>A</td><td>4</td><td>10</td></tr><tr><td>B</td><td>3</td><td>8</td></tr><tr><td>C</td><td>2</td><td>6</td></tr><tr><td>D</td><td>5</td><td>12</td></tr></table>	Job	Processing time(hr)	Due date(hr)	A	4	10	B	3	8	C	2	6	D	5	12	2	2	4			
Job	Processing time(hr)	Due date(hr)																				
A	4	10																				
B	3	8																				
C	2	6																				
D	5	12																				
Part –B																						
1	Explain how the Theory of Constraints can be applied to manage bottlenecks in a manufacturing environment. Provide examples of how identifying and addressing bottlenecks can improve overall production efficiency.	10	3	3																		
2	Consider the following flow shop scheduling problem with two machines (Machine 1 and Machine 2) and five jobs. The processing times (in hours) for each job on the two machines are as follows: <table><tr><th>Job</th><th>Machine 1</th><th>Machine 2</th></tr><tr><td>A</td><td>5</td><td>4</td></tr><tr><td>B</td><td>7</td><td>6</td></tr><tr><td>C</td><td>3</td><td>5</td></tr><tr><td>D</td><td>4</td><td>3</td></tr><tr><td>E</td><td>6</td><td>7</td></tr></table>	Job	Machine 1	Machine 2	A	5	4	B	7	6	C	3	5	D	4	3	E	6	7	10	3	4
Job	Machine 1	Machine 2																				
A	5	4																				
B	7	6																				
C	3	5																				
D	4	3																				
E	6	7																				

**Academic year 2024-2025 (ODD Sem)**

Using Johnson's Rule, find the optimal job sequence that minimizes the makespan.

3 Consider the following 3 machines and 5 jobs flow shop problem.

Job, i	Processing Time		
	Machine 1	Machine 2	Machine 3
1	10	10	12
2	12	8	20
3	16	6	14
4	12	4	20
5	20	8	8

Check whether the Johnson's algorithm can be extended to this problem. If yes, find the optimal sequence and the corresponding makespan.

4 Use graphical method to minimize the time needed to process the following jobs on the machines shown (i.e. for each machine, find the job which should be scheduled first). Also, calculate the total time elapsed to complete both jobs.

Job 1	Sequence	A	B	C	D	E
	Time (Hrs).	4	3	6	2	7
Job 2	Sequence	C	B	E	D	A
	Time (Hrs)	6	3	5	3	7

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4
	Quiz	Max Marks	-	-	4	6	4	6	-	-
	Test		-	-	10	40	-	--	35	15



Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	28 th January 2025	Maximum Marks	50
Course Code	IM3531A	Duration	90 Min
Sem	V	Improvement CIE	
QUALITY ASSURANCE			

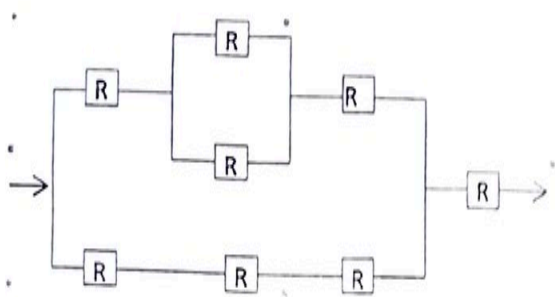
PART - A

Max Marks: 10

Sl No.	Questions	M	BT	CO
1	Three components each with a reliability of 0.9 are placed in series. What is the reliability of the system?	2	L2	CO1
2	There are two components that are identical in a parallel configuration where the second unit is in standby and will be activated only upon failure of the primary unit. The switching to the backup unit is considered certain. The failure rate for each component is 0.002 failures/hour. What is the reliability for 200 hours?	2	L1	CO2
3	The failure rates for two units in a parallel configuration, where the secondary unit is on back up until switched on upon failure of the primary unit, are each 0.002 failures/hour and the switch reliability is 0.95 per operation. What is the reliability for 300 hours?	2	L2	CO3
4	List the Advantages of CRD	2	L2	CO3
5	Expand ANOVA and ANCOVA	2	L2	CO2

PART - B

Max Marks: 50

Sl No	Questions	M	CO	BT
1	Define the following terms: Reliability; MTTF; MTBF; MTTR; Availability, Maintainability	10	CO3	3
2	Explain the utility of 'bath tub curve'. What are the reasons for failure during the three stages of equipment life?	10	CO1	3
3	Consider the following series-parallel configuration: 	10	CO2	3
Compute the System reliability for $R = 0.96$.				
4	List the General practical steps and guidelines for planning and conducting DOE	10	CO4	4



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5.a	How do the principles of replication and randomization work together to strengthen the internal validity of an experiment?	5	CO2
5.b	What are the key assumptions behind the application of a full factorial design and how do they affect the interpretation of experimental results?	5	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4
	Max Marks	Quiz	02	04	04	--	--	02	08	--	--
		Test	10	20	10	10	--	--	30	10	10



Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	28th January 2025	Maximum Marks	10 + 50
Course Code	IM254TA	Duration	120 Mins
Sem	V Semester	Improvement Test	
FINANCIAL ACCOUNTING AND COSTING			

Note:

1. Answer all the Questions.

PART - A

Sl. No.	Questions	M	BT	CO
1	Mention the main purpose of a Profit and Loss Account?	02	2	CO3
2	List the two main sections of a Balance Sheet?	02	1	CO3
3	Differentiate between Cost of Production and Cost of Sales?	02	2	CO1
4	What are the main components of Batch Costing?	02	1	CO4
5	What is the meaning of 'Equivalent Units' in process costing?	02	2	CO4

PART - B

Sl. No.	Questions	M	BT	CO																						
1.	Prepare a Trading and Profit and Loss Account for the year ending 31 st December 2024 from the following information. Also interpret the financial status of the organization. <table><tr><th>Particulars</th><th>Amount (₹)</th></tr><tr><td>Opening Stock</td><td>1,00,000</td></tr><tr><td>Purchases</td><td>5,00,000</td></tr><tr><td>Sales</td><td>8,00,000</td></tr><tr><td>Wages</td><td>50,000</td></tr><tr><td>Carriage Inwards</td><td>20,000</td></tr><tr><td>Salaries</td><td>60,000</td></tr><tr><td>Office Rent</td><td>25,000</td></tr><tr><td>Depreciation on Furniture (10%)</td><td>15,000</td></tr><tr><td>Interest on Loan</td><td>10,000</td></tr><tr><td>Closing Stock</td><td>1,50,000</td></tr></table>	Particulars	Amount (₹)	Opening Stock	1,00,000	Purchases	5,00,000	Sales	8,00,000	Wages	50,000	Carriage Inwards	20,000	Salaries	60,000	Office Rent	25,000	Depreciation on Furniture (10%)	15,000	Interest on Loan	10,000	Closing Stock	1,50,000	10	4	CO3
Particulars	Amount (₹)																									
Opening Stock	1,00,000																									
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Office Rent	25,000																									
Depreciation on Furniture (10%)	15,000																									
Interest on Loan	10,000																									
Closing Stock	1,50,000																									
2.	Prepare a Balance Sheet for ABC & Co. as on 31 st December 2024 using the following details: - Capital: ₹3,00,000 - Profit for the Year: ₹50,000 - Machinery: ₹2,50,000 - Creditors: ₹1,00,000 - Cash in Hand: ₹20,000 - Debtors: ₹1,00,000 - Bank Loan: ₹50,000 - Stock: ₹1,30,000	10	3	CO3																						



Academic year 2024-2025 (ODD Sem)

3.	Prepare a Cost Sheet for XYZ Ltd. for the year ending 31st March 2024 from the following data: - Raw Material Purchased: ₹2,00,000 - Freight on Raw Materials: ₹20,000 - Direct Labor: ₹1,50,000 - Direct Expenses: ₹25,000 - Factory Overheads: ₹1,00,000 (60% of Direct Labor) - Office Overheads: ₹75,000 (10% of Factory Cost) - Selling Overheads: ₹50,000 - Distribution Overheads: ₹30,000 - Profit: ₹1,20,000	10	3
4.	XYZ Ltd. is working on Job No. 45. The following details are provided for the job: - Direct Materials Used: ₹50,000 - Direct Wages: ₹30,000 - Machine Hours Worked: 20 hours - Machine Hour Rate: ₹500 per hour - Office Overheads: 20% of Prime Cost - Profit Markup: 25% on Total Cost Prepare a Job Cost Sheet for Job No. 45.	10	3
5.	XYZ Ltd. uses process costing to manufacture goods that go through three processes: A, B, and C. The following information is for Process B for the month of January 2025: - Opening Work-in-Progress (WIP): 500 units valued at ₹10,000 (Material: ₹20,000; Labor: ₹5,000; Overheads: ₹5,000) - Raw Materials Added: ₹40,000 - Direct Labor: ₹25,000 - Factory Overheads: ₹12,500 - Units Introduced from Process A: 5,000 - Units Transferred to Process C: 4,800 - Closing WIP: 700 units (Material 100% complete, Conversion 50% complete) Prepare the Cost Sheet for Process B.	10	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks											
Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4
	Max Marks	Quiz	02	--	04	04	--	02	--	04	04
		Test	--	--	20	30	--	--	--	40	10



Academic Year 2024-2025 (ODD Sem)

**DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT**

Date	29 th January 2025	Maximum Marks	50
Course Code	IM355TBD	Duration	90 Min
Sem	V	CIE – Improvement	
ADVANCED DECISION MODELLING (Professional Core Elective)			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
Part A				
1.	List two functional roles of inventory.	02	1	1
2.	Identify two key factors in inventory problem analysis.	02	2	3
3.	How does an EOQ model without shortages work?	02	2	2
4.	How does inventory contribute to production efficiency?	02	2	2
5.	How is Markov Analysis used in business decision-making?	02	2	3
Part B				
1.	The production department of a company requires 3,600 kg of raw material for manufacturing a particular item per year. It has been estimated that the cost of placing an order is Rs 36 and the cost of carrying inventory is 25 per cent of the investment in the inventories. The price is Rs 10 per kg. Help the purchase manager to determine an ordering policy for raw material.	10	1	1
2.	Each unit of an item costs a company Rs 40. Annual holding costs are 18 per cent of unit cost of the item due to miscellaneous changes: 1 per cent for insurance, 2 per cent allowances for obsolescence, Rs 2 for building overheads, Rs 1.50 for damage and loss, and Rs 4 miscellaneous costs. The annual demand for the item is constant at 1,000 units. Placing each order costs the company Rs 100. i. Calculate EOQ and the total costs associated with stocking the item. ii. If the supplier of the item will only deliver batches of 250 units, how are the stock holding costs affected? iii. If the supplier relaxes his order size requirement, but the company has limited warehouse space and can stock a maximum of 100 units at any time, what would be the optimal ordering policy and associated costs?	10	2	2
3.	A product is sold at the rate of 50 pieces per day and is manufactured at a rate of 250 pieces per day. The set-up cost of the machines is Rs 2,000 and the storage cost is found to be Re 0.15 per piece per day. With labour charges of Rs 3.20 per piece, material cost at Rs 2.10 per piece and overhead cost of Rs 4.10 per piece, find the minimum cost batch size if the interest charges are 8 per cent (assume 300 working days in a year). Compute the optimal number of cycles required in a year for manufacturing of this product.	10	3	3
4.	A commodity is to be supplied at a constant rate of 200 units per day. Supplies of any amount can be obtained at any required time, but each ordering costs Rs 50; cost of holding the commodity in inventory is Rs 2 per unit per day while the delay in the supply of the item induces a penalty of Rs 10 per unit per day. Find the optimal policy (Q, t), where t is the reorder cycle period and Q is the	10	3	3



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Academic Year 2024-2025 (ODD Sem)

	inventory after reorder. What would be the best policy to adopt if the penalty cost becomes infinite?		
5.	List and explain the characteristics of Markov Chain	10	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution		Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5
	Quiz	Max Marks	02	04	04	--	02	08	--	--	--
	Test	Max Marks	10	20	20	--	10	20	20	--	--



Academic year 2024-2025(Odd Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	29 th January 2025	Maximum Marks	10+50
Course Code	IM255TBB	Duration	120 Min
Sem	V Semester	Improvement CIE	
ENTERPRISE INFORMATION SYSTEMS			

Note:

1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	Define Business Process Automation (BPA).	02	L2	CO3
2	Name two common tools used in Robotic Process Automation (RPA).	02	L2	CO3
3	Mention one advantage of implementing automation in business processes.	02	L1	CO2
4	What is a key characteristic of structured data?	02	L1	CO1
5	What are the three types of data analytics?	02	L2	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	A mid-sized manufacturing company faces delays in order processing due to repetitive manual tasks such as data entry and validation. Propose a solution using Robotic Process Automation (RPA). Discuss how implementing RPA can streamline the process and improve efficiency.	10	L3	CO2
2.	A retail company plans to automate its invoice processing system using RPA. Explain the steps involved in the RPA implementation process for this scenario.	10	L3	CO3
3.	Design a simplified approach to integrating Business Process Automation (BPA) in the inventory management process of an e-commerce company. Include key steps and expected outcomes.	10	L4	CO4
4.	A retail chain plans to improve its inventory management system using data warehousing. Explain how a data warehouse can help in managing inventory more effectively. Include key components of a data warehouse and their roles.	10	L3	CO3
5.	Explain how data analytics in Enterprise Information Systems (EIS) can be used to improve the decision-making process for a manufacturing company aiming to reduce operational costs.	10	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz Marks	02	04	04	--	04	06	--	--	--	--
	Test Marks	--	10	20	20	--	10	30	10	--	--